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Q1: What are chromosomal aberration, how do we classify them. 10

A chromosome anomaly, abnormality, aberration, or mutation is a missing, extra, or irregular portion of chromosomal DNA.[1] It can be from an atypical number of chromosomes or a structural abnormality in one or more chromosomes. Chromosome mutation was formerly used in a strict sense to mean a change in a chromosomal segment, involving more than one gene.[2] A karyotype refers to a full set of chromosomes from an individual that can be compared to a "normal" karyotype for the species via genetic testing. A chromosome anomaly may be detected or confirmed in this manner. Chromosome anomalies usually occur when there is an error in cell division following meiosis or mitosis. There are many types of chromosome anomalies. They can be organized into two basic groups, numerical and structural anomalies.

Types:

The three major single chromosome mutations; deletion (1), duplication (2) and inversion (3). The two major two chromosome mutations; insertion (1) and Translocation (2). A chromosome anomaly, abnormality, aberration, or mutation is a missing, extra, or irregular portion of chromosomal DNA.

Q2: Point mutation? Revers forward neutral mutation? 5 marks

A point mutation, or single base modification, is a type of mutation that causes a single nucleotide base substitution, insertion, or deletion of the genetic material, DNA or RNA. The term frame shift mutation indicates the addition or deletion of a base pair.

Point mutation is a random SNP (single-nucleotide polymorphism) mutation in the deoxyribonucleic acid (DNA) that occurs at one point. Point mutations usually take place during DNA replication. DNA replication occurs when one double-stranded DNA molecule creates two single strands of DNA, each of which is a template for the creation of the complementary strand. A single point mutation can change the whole DNA sequence. Changing one purine or pyrimidine may change the amino acid that the nucleotides code for.

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Point mutations may arise from spontaneous mutations that occur during DNA replication. The rate of mutation may be increased by mutagens. Mutagens can be physical, such as radiation from UV rays, X-rays or extreme heat, or chemical (molecules that misplace base pairs or disrupt the helical shape of DNA). Mutagens associated with cancers are often studied to learn about cancer and its prevention.

Q3: What is epigenetics?

The study of changes in organisms caused by modification of gene expression rather than alteration of the genetic code itself.

Q5:.Write the modification of epigenetics? 10

Epigenetic modifications are crucial for packaging and interpreting the genome under the influence of physiological factors [4-8]. Epigenetics is one of the fastest-growing areas of science and has now become a central issue in biological studies of development and disease [3, 7-11]. In recent years, there have been rapid advancees in the understanding of epigenetic mechanisms [12-14], which include histone modifications [4-6, 15], DNA methylation [10, 11, 16, 18], small and non-coding RNAs [20, 56], and chromatin architecture [13, 15, 21]. These mechanisms, in addition to other transcriptional regulationary events [15, 17], ultimately regulate gene activity and expression during development and differentiation, or in response to environmental stimuli [16, 17].

Epigenetic research can help explain how cells carrying identical DNA differentiate into different cell types, and how they maintain differentiated cellular states [14, 17]. Epigenetics is thus considered a bridge between genotype and phenotype [1-3, 7, 9].

While epigenetics refers to the changes of single genes or sets of genes, the term epigenome reflects the overall epigenetic state of a cell, and refers to global analyses of epigenetic markers across the entire genome [3, 14]. It is therefore critically important to map the epigenetic

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modification patterns or profile the epigenome in a given cell, which then can be used as epigenetic biomarkers for clinical prediction, diagnosis, and therapeutic development [8, 11, 18, 45]. International human epigenome projects are currently working to catalog all the epigenetic markers in all major tissues across the entire genome. The resulting reference maps will usher in epigenetics as an exciting new era of medical science [14]. As an example of the research community's commitment to classifying epigenetic markers, the National Institutes of Health (NIH) has recently launched a $190 million research effort to learn more about epigenetics. This mini-review focuses on DNA methylation and the predominant histone modifications, with emphasis on their dynamic interactions within the chromatin environment to form the complex epigenetic mechanisms that orchestrate the regulation of genes at the molecular level in mammalian cells.

Q6: RNA editing explanation?

RNA editing is a molecular process through which some cells can make discrete changes to specific nucleotide sequences within a RNA molecule after it has been generated by RNA polymerase. RNA editing is relatively rare, and common forms of RNA processing (e.g. splicing, 5'-capping and 3'-polyadenylation) are not usually included as editing. Editing events may include the insertion, deletion, and base substitution of nucleotides within the edited RNA molecule.

RNA editing has been observed in some tRNA, rRNA, mRNA or miRNA molecules of eukaryotes and their viruses, archaea and prokaryotes.[1] RNA editing occurs in the cell nucleus and cytosol, as well as within mitochondria and plastids. In vertebrates, editing is rare and usually consists of a small number of changes to the sequence of affected molecules. In other organisms, extensive editing (pan-editing) can occur; in some cases the majority of nucleotides in a mRNA sequence may result from editing.

RNA-editing processes show great molecular diversity, and some appear to be evolutionarily

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recent achievements that arose independently. The diversity of RNA editing phenomena includes nucleobase modifications such as cytidine (C) to uridine (U) and adenosine (A) to inosine (I) deaminations, as well as non-templated nucleotide additions and insertions. RNA editing in mRNAs effectively alters the amino acid sequence of the encoded protein so that it differs from that predicted by the genomic DNA sequenc

Editing by insertion or deletion

RNA editing through the addition and deletion of uracil has been found in kinetoplasts from the mitochondria of Trypanosomabrucei[3] Because this may involve a large fraction of the sites in a gene, it is sometimes called "pan-editing" to distinguish it from topical editing of one or a few sites.

Pan-editing starts with the base-pairing of the unedited primary transcript with a guide RNA (gRNA), which contains complementary sequences to the regions around the insertion/deletion points. The newly formed double-stranded region is then enveloped by an editosome, a large multi-protein complex that catalyzes the editing.[4][5] The editosome opens the transcript at the first mismatched nucleotide and starts inserting uridines. The inserted uridines will base-pair with the guide RNA, and insertion will continue as long as A or G is present in the guide RNA and will stop when a C or U is encountered.[6][7] The inserted nucleotides cause a frameshift and result in a translated protein that differs from its gene.

The Effect of Uracil Insertion in pre-mRNA transcripts

The mechanism of the editosome involves an endonucleolytic cut at the mismatch point between the guide RNA and the unedited transcript. The next step is catalyzed by one of the enzymes in the complex, a terminal U-transferase, which adds Us from UTP at the 3’ end of the mRNA.[8] The opened ends are held in place by other proteins in the complex. Another enzyme, a U-specific exoribonuclease, removes the unpaired Us. After editing has made mRNA complementary to gRNA, an RNA ligase rejoins the ends of the edited mRNA transcript.[9][10] As a consequence, the editosome can edit only in a 3’ to 5’ direction along the primary RNA

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transcript. The complex can act on only a single guide RNA at a time. Therefore, a RNA transcript requiring extensive editing will need more than one guide RNA and editosome complex

Q7: What is CRISPRs? Define its work and application.

Clustered regularly interspaced short palindromic repeats (CRISPR, pronounced crisper[2]) are segments of prokaryoticDNA containing short, repetitive base sequences. In a palindromic repeat, the sequence of nucleotides is the same in both directions. Each repetition is followed by short segments of spacer DNA from previous exposures to foreign DNA (e.g., a virus or plasmid).[3] Small clusters of cas (CRISPR-associated system) genes are located next to CRISPR sequences.

The CRISPR/Cas system is a prokaryotic immune system that confers resistance to foreign genetic elements such as those present within plasmids and phages[4][5][6] that provides a form of acquired immunity. RNA covering the spacer sequence helps Cas proteins recognize and cut exogenous DNA. Other RNA-guided Cas proteins cut foreign RNA.[7] CRISPRs are found in approximately 40% of sequenced bacterial genomes and 90% of sequenced archaea.[8][note 1] A simple version of the CRISPR/Cas system, CRISPR/Cas9, has been modified to edit genomes. By delivering the Cas9 nuclease complexed with a synthetic guide RNA (gRNA) into a cell, the cell's genome can be cut at a desired location, allowing existing genes to be removed and/or new ones added.[9][10][11] The Cas9-gRNA complex corresponds with the CAS III crRNA complex in the above diagram.

CRISPR/Cas genome editing techniques have many potential applications, including medicine and crop seed enhancement. The use of CRISPR/Cas9-gRNA complex for genome editing[12][13] was the AAAS's choice for breakthrough of the year in 2015.[14]Bioethical concerns have been raised about the prospect of using CRISPR for germline editing.[15]

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Q8: What are simple lipid? Give two types of simple lipid??

A simple lipid is a fatty acid ester of different alcohols and carries no other substance. These lipids belong to a heterogeneous class of predominantly nonpolar compounds, mostly insoluble in water, but soluble in nonpolar organic solvents such as chloroform and benzene. example: fats and oils.

Q9: Name three functional group of purines and pyrimidines??

Nitro group

Exocyclic amino group

Carbonyl group

Q10: Two example of unnatural pyrimidine bases??

Fluorouracil or Azc

Q11: Difference between nucleotide and nucleoside??

In medicine several nucleoside analogues are used as antiviral or anticancer agents

Topical application of nucleotides increase lysozyme levels in tears

Q12: What is waxes? Explain in detail.

Waxes are a diverse class of organic compounds that are hydrophobic, malleable solids near ambient temperatures. They include higher alkanes and lipids, typically with melting points above about 40 °C (104 °F), melting to give low viscosity liquids. Waxes are insoluble in water but soluble in organic, nonpolar solvents. Natural waxes of different types are produced by plants and animals and occur in petroleum.

Animal waxes[edit]

The most commonly known animal wax is beeswax, but other insects secrete (release) waxes. A

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major component of the beeswax used in constructing honeycombs is the ester myricyl palmitate which is an ester of triacontanol and palmitic acid. Its melting point is 62-65 °C. Spermaceti occurs in large amounts in the head oil of the sperm whale. One of its main constituents is cetyl palmitate, another ester of a fatty acid and a fatty alcohol. Lanolin is a wax obtained from wool, consisting of esters of sterols.[1]

Plant waxes[edit]

Plants secrete waxes into and on the surface of their cuticles as a way to control evaporation, wettability and hydration.[3] The epicuticular waxes of plants are mixtures of substituted long-chain aliphatic hydrocarbons, containing alkanes, alkyl esters, fatty acids, primary and secondary alcohols, diols, ketones, aldehydes.[2] From the commercial perspective, the most important plant wax is carnauba wax, a hard wax obtained from the Brazilian palm Copernicia prunifera. Containing the ester myricyl cerotate, it has many applications, such as confectionery and other food coatings, car and furniture polish, floss coating, surfboard wax and other uses. Other more specialized vegetable waxes include candelilla wax and ouricury wax.

Q13: What is enzyme exp?

A substance produced by a living organism which acts as a catalyst to bring about a specific biochemical reaction. Examples are lactase, alcohol dehydrogenase and DNA polymerase.

Q14: Give example of condensation.

Examples of Condensation: 1. Having a cold soda on a hot day, the can "sweats." Water molecules in the air as a vapor hit the colder surface of the can and turn into liquid water.

Q15: Two properties of glyceroltrinitrate - 2

It is soluble 1 in 800 of water and 1 in 4 of ethanol. It is miscible with acetone, glacial acetic acid, ethyl acetate, benzene, pyridine, chloroform, chloroform and ether. It is sparingly soluble

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in petroleum ether, liquid petrolatum and glycerol.

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